

Mobile and Embedded Computing

Lecture 5. GraphQL & REST, Observability for Mobile applications (Events, Crashlytics, Performance Monitoring)

Section 1

Mobile API Communication

Why APIs are Critical

-  **Data Source:** Apps are often containers; the data lives on remote servers.
-  **Functionality:** Enable complex features like login, payments, and social feeds.
-  **Synchronization:** Keep user data consistent across multiple devices (phone, tablet, web).
-  **Separation of Concerns:** Decouples the client (app) from the backend (server).

REST API Fundamentals

Architecture

REpresentational **S**tate **T**ransfer. A stateless, client-server architectural style.

It is not a protocol, but a set of constraints.

Core Concepts

Uses standard HTTP methods to act on "Resources" identified by URLs.

GET /users/123

POST /users

PUT /users/123

DELETE /users/123

REST in Mobile: Pros & Cons



Pros

Simplicity: Easy to understand, built on standard HTTP.

Caching: Standard HTTP caching is effective.

Mature: Decades of tooling and support.



Cons

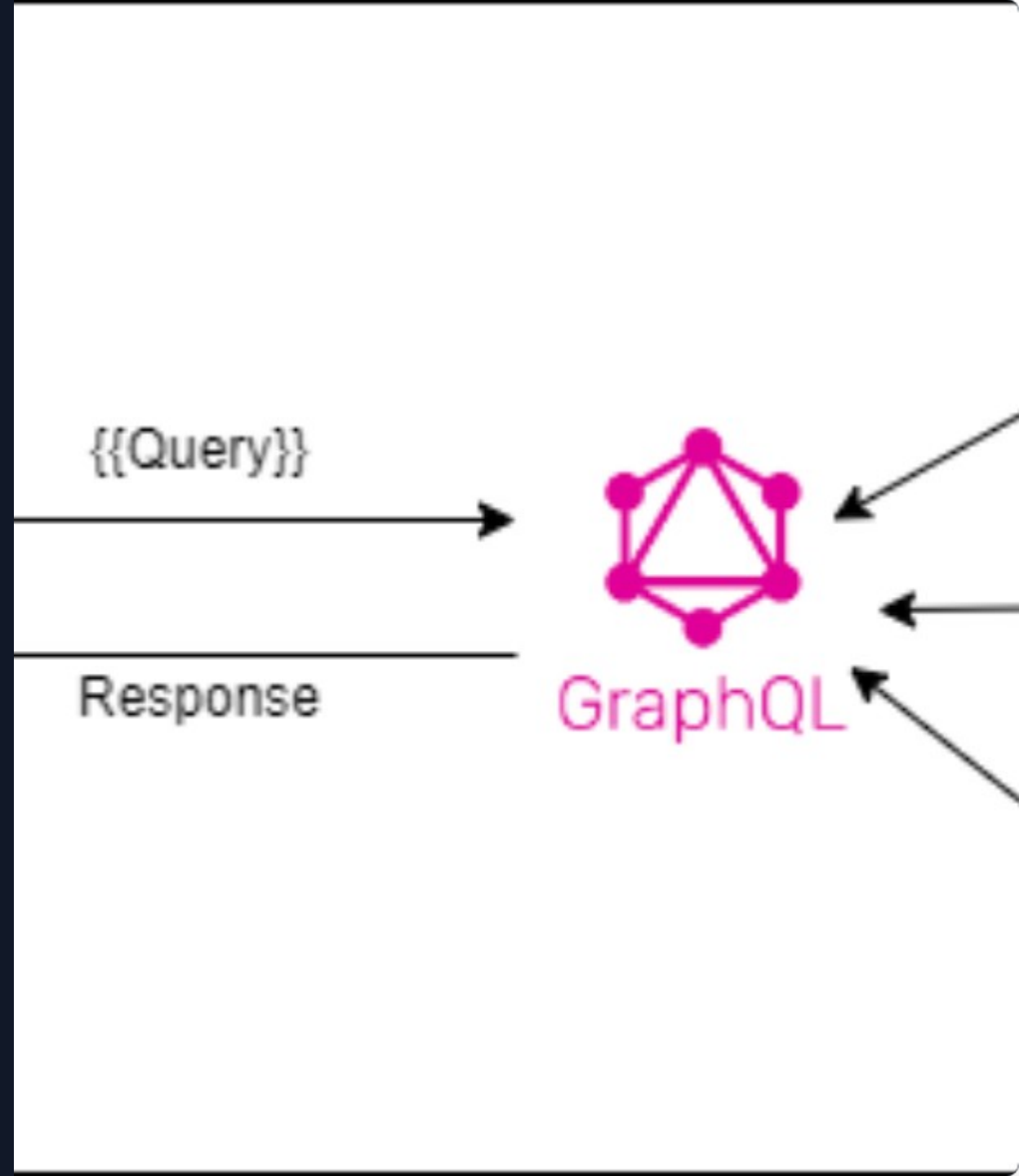
Over-fetching: Getting more data than the screen needs.

Under-fetching: Needing multiple API calls for one screen (e.g., user + posts).

Introduction to GraphQL

A **query language** for your API, not an architecture. Developed by Facebook to solve mobile data-fetching issues.

Allows the client to ask for *exactly* what it needs and get *exactly* that in a single request.



GraphQL Core Concepts



Schema

The "contract." A strongly-typed definition of all data available on the API (types, fields, relationships).



Query

A read-only operation to fetch data.
The query's shape mirrors the JSON response's shape.



Mutation

A write operation to modify data (create, update, delete). Defined in the schema just like queries.

GraphQL vs. REST: Head-to-Head

Feature	REST	GraphQL
Endpoint	Multiple endpoints (/users, /posts)	Single endpoint (e.g., /graphql)
Data Fetching	Fixed data structure (defined by server)	Flexible data shape (defined by client)
Data Shape	Prone to over/under-fetching	Precise data in one request
Schema	Loosely defined (e.g., OpenAPI)	Strongly typed (Schema Definition Language)
Protocol	Uses all HTTP verbs (GET, POST, PUT...)	Typically uses POST for all operations

When to Use...

Use REST when...

- ...simplicity is key.
- ...you have simple, well-defined resources.
- ...caching is a top priority (HTTP caching works perfectly).
- ...you are integrating with many 3rd-party APIs.

Use GraphQL when...

- ...you have complex data relationships (e.g., social feeds).
- ...mobile bandwidth is a major concern (reduces payload).
- ...your app needs to fetch data from multiple sources.
- ...you have multiple clients (iOS, Android, Web) with different data needs.

Section 2

Mobile App Observability

What is Mobile Observability?



More than monitoring: Monitoring tells you **if** something is broken. Observability helps you understand **why**.



Three Pillars: Traditionally Logs, Metrics, and Traces.



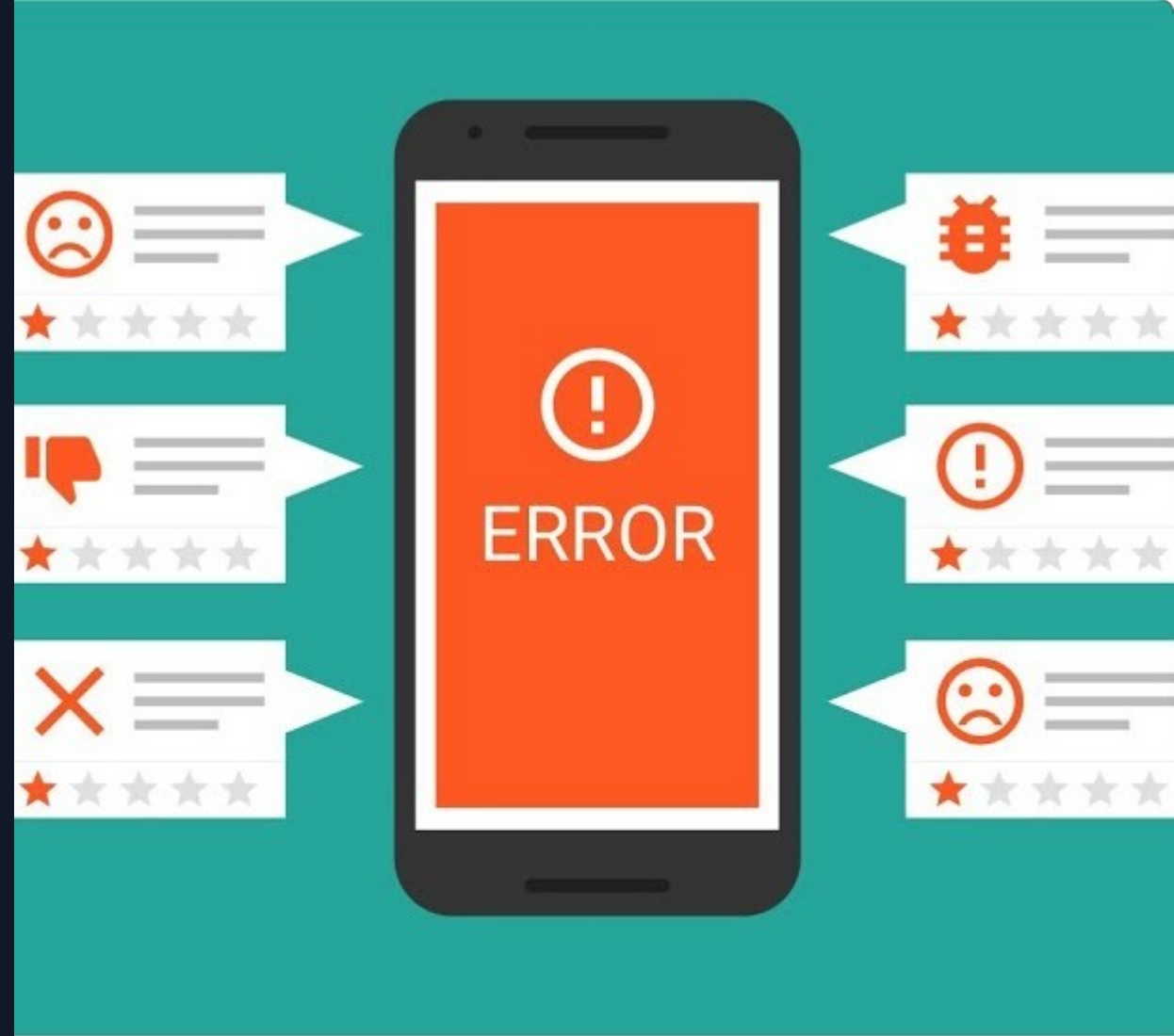
For Mobile: We adapt this to our "Big 3":

1. **Events** (User Behavior)
2. **Crashes** (Stability)
3. **Performance** (Speed)

The Firebase Toolkit

Google's mobile platform that provides a suite of tools for building, releasing, and monitoring apps.

We focus on its observability products, which work together to give a complete picture of app health.



Google Firebase Crashlytics

1. Firebase Analytics (Events)



What it is: Tracks user behavior and app usage by logging "Events".



Automatic Events: Captures screen_view, app_remove, first_open automatically.



Custom Events: You log events that matter to your app (e.g., purchase_complete, share_image).



The "Why": Understand user funnels, feature adoption, and engagement.

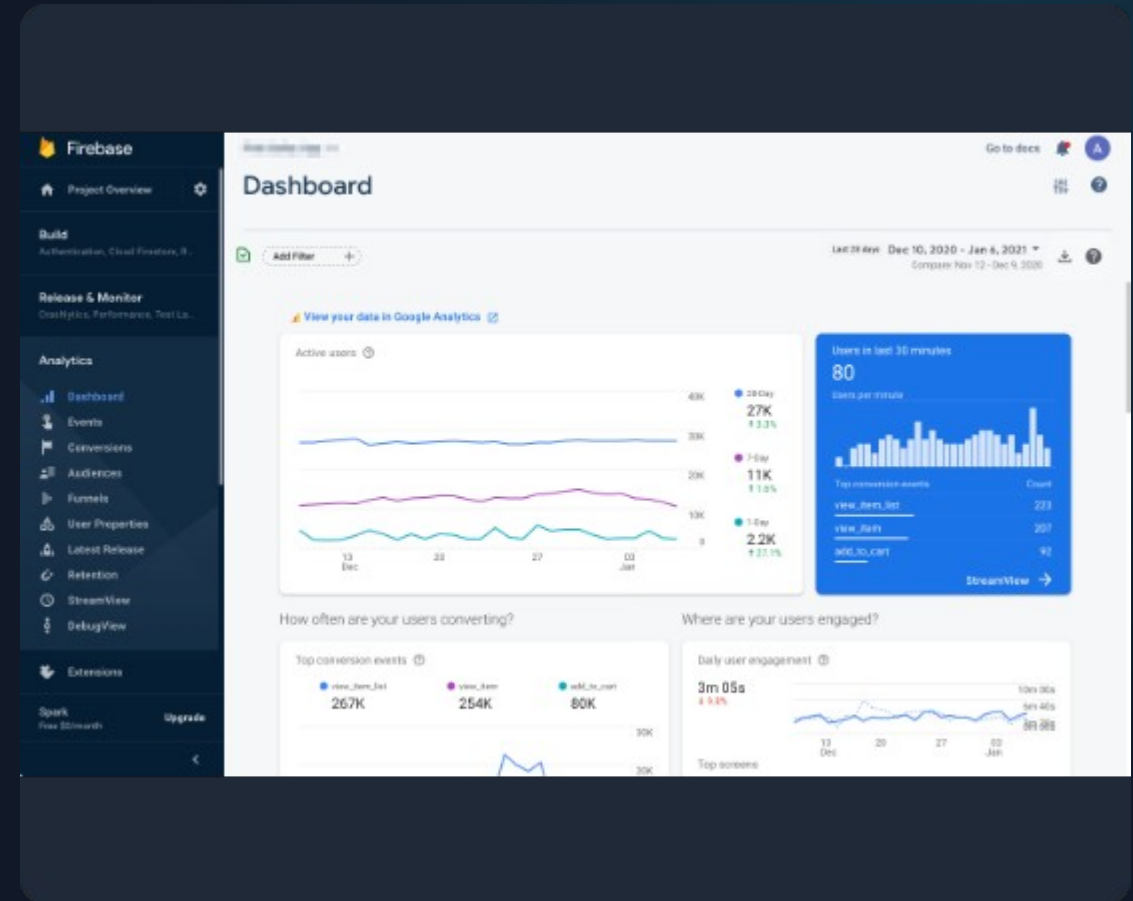
Implementing Firebase Events

1. Get the FirebaseAnalytics instance:

```
private FirebaseAnalytics analytics;  
analytics = FirebaseAnalytics.getInstance(this);
```

2. Log your custom event with a Bundle:

```
Bundle bundle = new Bundle();  
bundle.putString("item_name", "Red T-Shirt");  
bundle.putString("category", "Apparel");  
analytics.logEvent("select_content", bundle);
```



2. Firebase Crashlytics

99.9%
Goal: Crash-Free Users

Find, Fix, and Prioritize Crashes

Real-time crash reporting that automatically captures and groups crashes (Java/Kotlin & NDK).

Provides actionable insights: stack traces, device info, and OS version.

Setting Up Crashlytics



Integration: Add the Gradle dependency. It works out-of-the-box.



Custom Logs: Add "breadcrumb" context to reports.
`Crashlytics.log("User at checkout screen");`



Custom Keys: Set user state (e.g., premium/free).
`Crashlytics.setCustomKey("user_tier", "premium");`



Test Crash: Verify your setup.
`throw new RuntimeException("Test Crash");`

3. Firebase Performance



App Startup

Measures "cold," "warm," and "hot" app start times. A critical first impression.



Network Requests

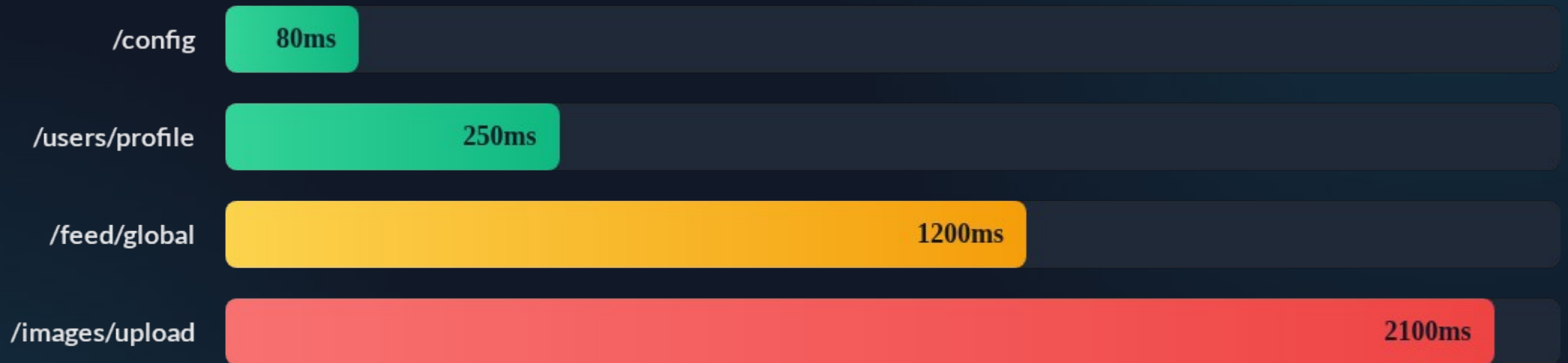
Automatically monitors HTTP/S requests. Tracks success rate, payload size, and response time.



Custom Traces

Manually trace specific code blocks (e.g., "image_processing") to find bottlenecks.

Performance Deep Dive: Network

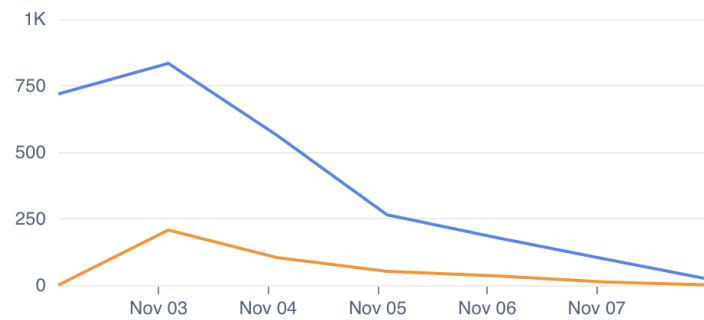


The /feed/global and /images/upload endpoints are clear bottlenecks that need investigation.

Total events by version

for the last 7 days

- 4.7.11 (436)
2.7K
- 4.7.15 (440)
415



Devices

63% Samsung

Operating systems

37% Android 15

Device states

2% background

63% Samsung →

13% Xiaomi →

8% Motorola →

5% Oppo →

11% Other (5)

Events



Alert your team when an important issue like this one occurs by setting up an integration

See integrations



Event summary

4.7.11 (436)

16

Galaxy S24 Ultra

Nov 9, 2025, 10:32:39 AM

Stack trace

Keys

Logs & Breadcrumbs

Data

Rollouts



TXT



Fatal Exception: io.flutter.plugins.firebase.crashlytics.FlutterError
Provider<ConsentsCubit> not found for ScannerAlimente



```
Fatal Exception: io.flutter.plugins.firebase.crashlytics.FlutterError: Provider<ConsentsCubit> not found for ScannerAlimente
at Provider._inheritedElementOf(provider.dart:377)
at Provider.of(provider.dart:327)
at ReadContext.read(provider.dart:683)
at _ScannerAlimenteState._checkAndPromptConsents(main.dart:657)
at _ScannerAlimenteState._initApp.<fn>(main.dart:526)
at SchedulerBinding._invokeFrameCallback(binding.dart:1434)
at SchedulerBinding.handleDrawFrame(binding.dart:1361)
at SchedulerBinding._handleDrawFrame(binding.dart:1200)
```

Event summary

4.7.11 (436)

16

Galaxy S24 Ultra

Nov 9, 2025, 10:32:39 AM

<

>

Stack trace

Keys

Logs & Breadcrumbs

Data

Rollouts

All

Logs

Events

Filter logs

Expand all

Download logs

#	Time (newest first) ↑	Source	Log
▶ 1	10:32:38.383 AM		app_open
▶ 2	10:32:38.160 AM		screen_view
▶ 3	10:32:38.080 AM		session_start

Items per page:

25

1 – 3 of 3

<

>

Percentile
90%

Filter

Last 7 days
Nov 3 – Nov 9

App start time

App start time (1.30s) is **8% faster** compared to 6 days prior



All versions

4.7.31 (456)

Select version

1.30s

1.09s

—

View metric details

App start time

-8% 1.30s

Duration

find-codes-in-background

+84% 188 ms

Duration

vertex-ai-generation

-25% 6.96s

Success rate

scan.cloud.codingshadows.com/**

+8 97.22%

Response time

scan.cloud.codingshadows.com/**

-26% 1.76s

Select a metric

+

Distribution ?

5% - 95% ▾

67 ms - 3.77s

— Median - 314 ms

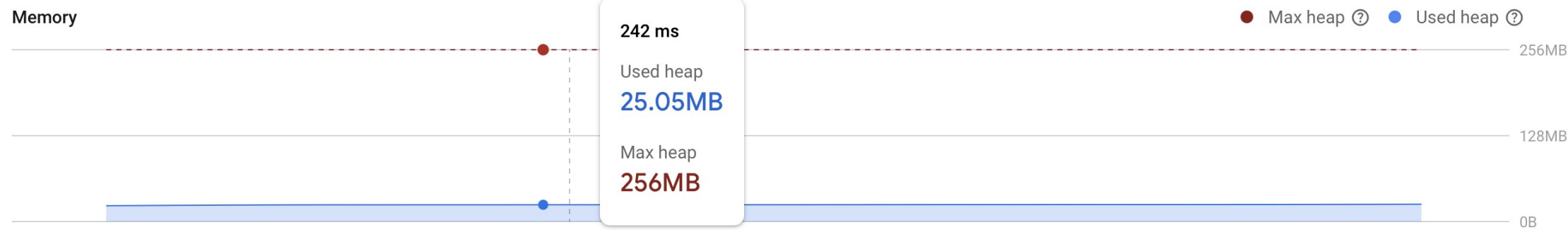


Sessions (9 of 811)



Start time	Response time	Request payload size	Response payload size	Response code	Response content type
Nov 8, 9:41:53 AM	282 ms	111B	542B	200	application/json
🔄 4.7.21 (446)	📱 Motorola moto g15 power	📶 35	🌐 Romania	📶 RCS&RDS Digi Mobile	📶 LTE
CPU ? 0s100 ms200 ms300 ms400 ms500 ms600 ms682 ms 241 ms Total CPU 109.8% System CPU 39.93% User CPU 69.87% ● User ? ● System ? 183.65% 91.83%					
Memory ● Max heap ? ● Used heap ?					

Memory



On device

product_prices_fetch

Network

scan.cloud.codingshadows.com...

scan.cloud.codingshadows.com/**

◆ Gemini

Requests
3.7K

Success rate
100%

Latency (p95)
3.61s

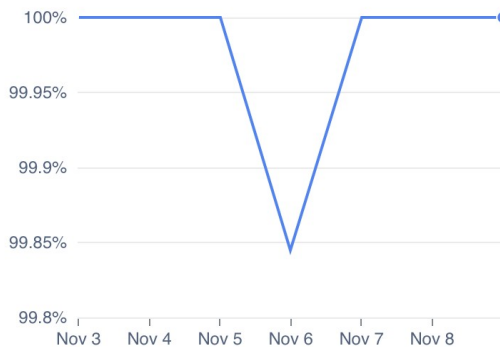
Tokens
6.8M



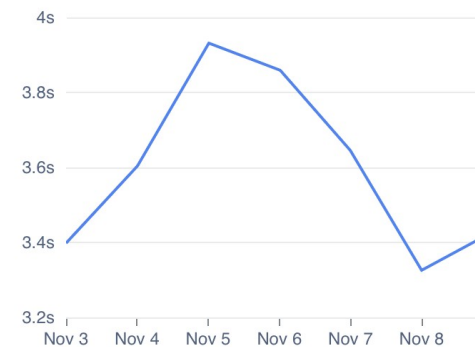
Requests



Success rate



Latency (p95)



Tokens

Text

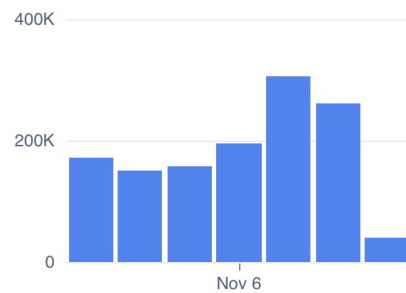
Image

Video

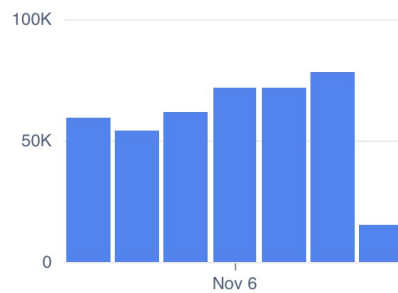
Audio

Document

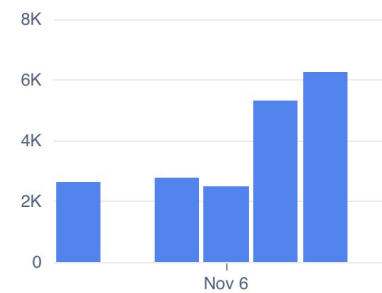
Input
1.3M



Output
412K



Thinking
18K



Traces

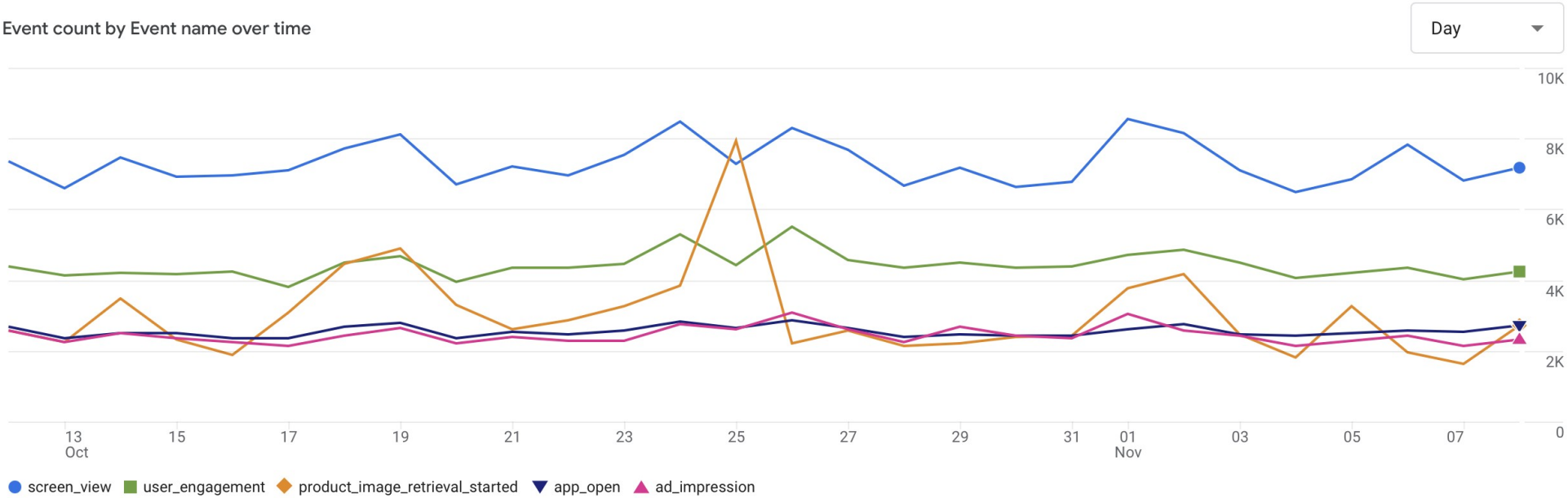
Sampling rate for all apps is 100%. [Edit rate](#) (?)

 Filter Add filter...

Status	Created	Input	Output	App version	Latency	Model
✔	Nov 9, 2025, 10:50:07 AM	{\"content\": {\"data\":\"image/jpeg<sha5121165cfe1ae3aa32f434063afe729054498d6700ff938d78c4c53dd92f2591431d097c56934e277c5f9fd9795ed...	{\"event.name\":\"gen_ai.choice\",\"finish_reason\":\"stop\",\"index\":0,\"message\": {\"content\":{\"text\":\"\\n...	unknown	1.56s	gemini-2.5-flash-lite
✔	Nov 9, 2025, 10:47:15 AM	{\"content\": {\"data\":\"image/jpeg<sha5128ee129e69ba6c8129e1e6b387e9357c4b7728e0c75f54f54da05f5751d914b3f0ea4d963a7365d9fa0f11ddb81...	{\"event.name\":\"gen_ai.choice\",\"finish_reason\":\"stop\",\"index\":0,\"message\": {\"content\":{\"text\":\"\\n...	unknown	1.61s	gemini-2.5-flash-lite
✔	Nov 9, 2025, 10:47:02 AM	{\"content\": {\"data\":\"image/jpeg<sha5120cb4ec7c4558b6a094afc145e85ddb7ea77d6269581aab2859b001f2bcd11755f9f7f8c33041ab8a9d88a44565...	{\"event.name\":\"gen_ai.choice\",\"finish_reason\":\"stop\",\"index\":0,\"message\": {\"content\":{\"text\":\"\\n...	unknown	2.01s	gemini-2.5-flash-lite
✔	Nov 9, 2025, 10:44:30 AM	{\"content\": {\"data\":\"image/jpeg<sha512cf3096b4cfb096bdbfa2dbe1ba637b02202f5925277be45842e082b721ea0b72e225f5505375917023e567f96...	{\"event.name\":\"gen_ai.choice\",\"finish_reason\":\"stop\",\"index\":0,\"message\": {\"content\":{\"text\":\"\\n...	unknown	1.53s	gemini-2.5-flash-lite
✔	Nov 9, 2025, 10:42:40 AM	{\"content\": {\"data\":\"image/jpeg<sha5123fbeda4f9ecbe80d737eae6203a1f96c83e4bbccdae4c30b516a08cf272d956990e55da0f94e1af7d0c30ba5439...	{\"event.name\":\"gen_ai.choice\",\"finish_reason\":\"stop\",\"index\":0,\"message\": {\"content\":{\"text\":\"lapte pasteurizat...	unknown	1.46s	gemini-2.5-flash-lite

Items per page: 5 ▼ ‹ ›

Event count by Event name over time



Plot rows

Search...

Rows per page: 10 ▾

Go to: 1 < 1-10 of 258 >

	Event name	Event count	Total users	Event count per active user	Total revenue	
☐	Total	1,471,901 100% of total	25,444 100% of total	72.16 Avg 0%	\$923.94 100% of total	
☒	1 screen_view	204,544 (13.9%)	21,018 (82.6%)	10.13	\$0.00 (0%)	⋮
☒	2 user_engagement	123,492 (8.39%)	19,990 (78.56%)	6.18	\$0.00 (0%)	⋮
☒	3 product_image_retrieval_started	84,300 (5.73%)	2,310 (9.08%)	36.83	\$0.00 (0%)	⋮
☒	4 app_open	71,411 (4.85%)	20,944 (82.31%)	3.54	\$0.00 (0%)	⋮

Real life example



1. Event

User clicks "Buy". (Analytics logs begin_checkout).



2. Performance

checkout_trace starts. Network call to /payment is **slow (5.2s)**. (Performance alerts).



3. Crash

App crashes due to an unexpected null response from the slow /payment API. (Crashlytics reports).